

Problem Statement

Evaluate the limit from the image "dailyintegral-limit-limit-1x1-2026-08-07.jpg":

$$\lim_{n \rightarrow \infty} n^{3/2} \left(\sqrt{n^3 + 1} - n\sqrt{n} \right)$$

Detailed Solution

To evaluate this limit, we can multiply the expression by its conjugate. This will eliminate the problematic square roots in the numerator and allow us to evaluate the limit as n approaches infinity by dividing by the highest power of n .

Step 1: Rewrite the Expression

First, let's rewrite the term $n\sqrt{n}$ as $\sqrt{n^3}$ so it matches the format of the other square root:

$$\lim_{n \rightarrow \infty} n^{3/2} \left(\sqrt{n^3 + 1} - \sqrt{n^3} \right)$$

Step 2: Multiply by the Conjugate

Multiply the numerator and denominator by the conjugate of the expression inside the parentheses, which is $(\sqrt{n^3 + 1} + \sqrt{n^3})$:

$$\lim_{n \rightarrow \infty} n^{3/2} \frac{(\sqrt{n^3 + 1} - \sqrt{n^3})(\sqrt{n^3 + 1} + \sqrt{n^3})}{\sqrt{n^3 + 1} + \sqrt{n^3}}$$

Step 3: Simplify the Numerator

Using the difference of squares formula, $(a-b)(a+b) = a^2 - b^2$, we can simplify the numerator:

$$\lim_{n \rightarrow \infty} n^{3/2} \frac{(n^3 + 1) - n^3}{\sqrt{n^3 + 1} + \sqrt{n^3}}$$

The n^3 terms cancel out perfectly, leaving us with:

$$\lim_{n \rightarrow \infty} \frac{n^{3/2}}{\sqrt{n^3 + 1} + \sqrt{n^3}}$$

Step 4: Divide by the Highest Power of n

To evaluate the limit as $n \rightarrow \infty$, we divide both the numerator and the denominator by $n^{3/2}$. Remember that dividing by $n^{3/2}$ outside a square root is equivalent to dividing by n^3 inside a square root:

$$\lim_{n \rightarrow \infty} \frac{\frac{n^{3/2}}{n^{3/2}}}{\frac{\sqrt{n^3+1}}{\sqrt{n^3}} + \frac{\sqrt{n^3}}{\sqrt{n^3}}}$$

Simplify the fractions:

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{\frac{n^3+1}{n^3}} + 1}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{n^3}} + 1}$$

Step 5: Evaluate the Limit

As n approaches infinity, the term $\frac{1}{n^3}$ approaches 0. Substituting this in allows us to evaluate the final value:

$$\frac{1}{\sqrt{1+0}+1} = \frac{1}{1+1} = \frac{1}{2}$$

Final Answer:

$$\frac{1}{2}$$